

A

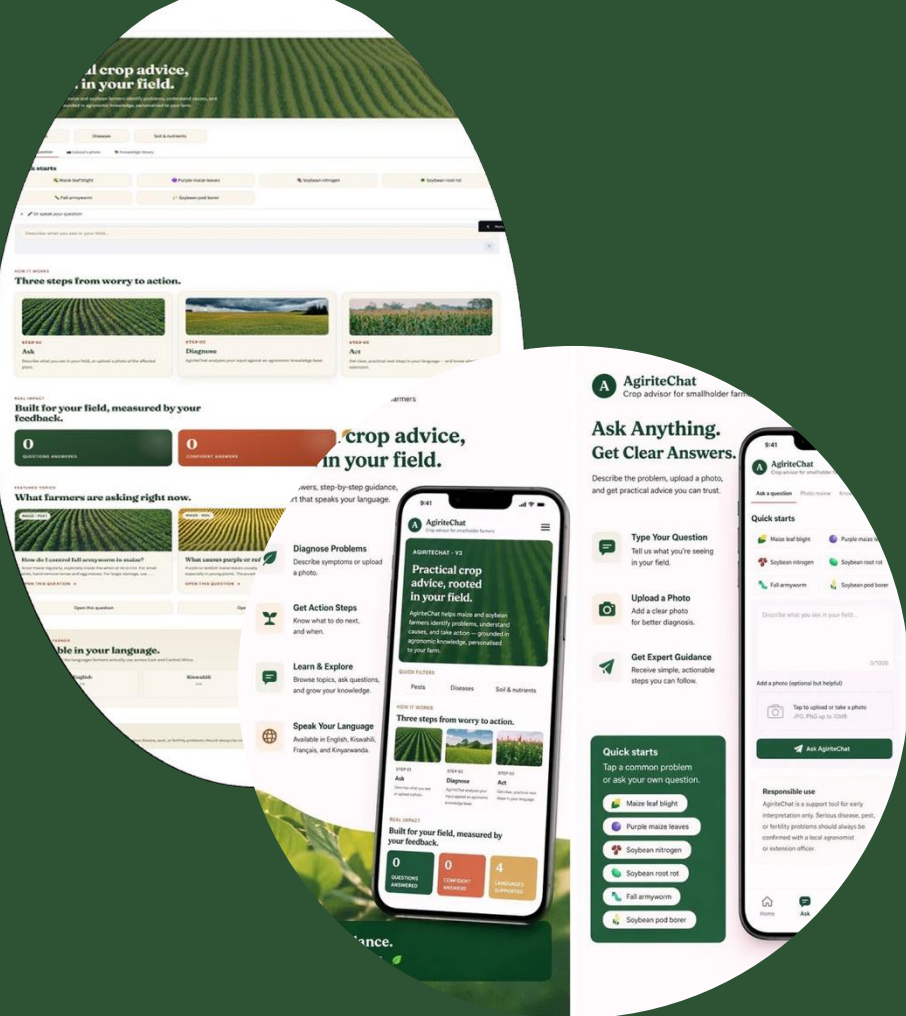
AgiriteChat

Agent-Based AI Decision Support
for Sustainable Agriculture

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agiritechat.streamlit.app

101 KB entries · 7-node agent · 4 languages · \$0 cost



THE PROBLEM

Farmers need advice. Experts are overstretched.

Small-scale farmers lack timely expert advice. Extension officers are overstretched. Existing AI tools hallucinate. Farmers need grounded, reliable guidance.

1 : 3,000

Extension officer
to farmer ratio

800 mm

Annual rainfall
Rwanda Eastern Province

30–40%

Post-harvest loss
without proper storage

Focus: Nebraska · Rwanda Eastern Province (Kayanza, Ngoma, Kirehe, Nyagatare, Bugesera)

7-Node LangGraph Agent

1

CLASSIFY

Greeting? Crop name? Real question?

2

RETRIEVE

Semantic search · cosine similarity

3

CONFIDENCE

Score ≥ 0.55 ? 0.35–0.55? Below?

4

SYNTHESIZE

Grounded 5-section advisory

!

REFUSE

"I don't know" → extension officer

CORE PRINCIPLE

An agent is defined by knowing when not to act.

The refusal path is the most important node in the system.

When confidence is low, the system refuses to guess and directs the farmer to a local extension officer.

In agriculture, silence is safer than a confident wrong answer.

FEATURES

What it does — end to end



Text Advisory

5-section grounded answers from 101-entry KB



Photo Diagnosis

Vision extracts symptoms, KB diagnoses



Voice Input

Whisper transcribes, agent answers



4 Languages

EN, SW, FR, RW — native generation



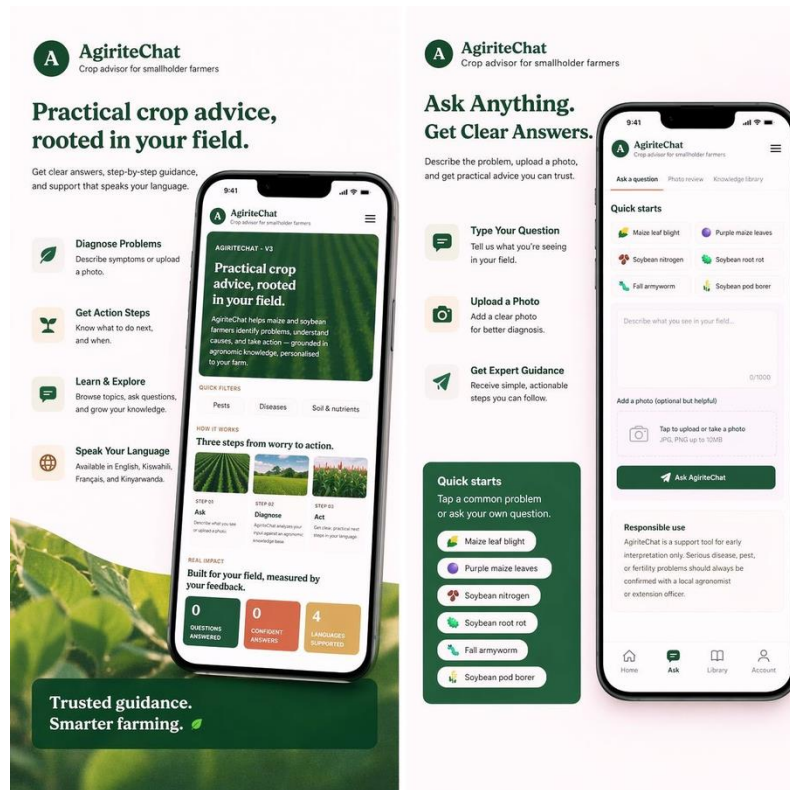
Weather

Current + 3-day forecast



Source Separation

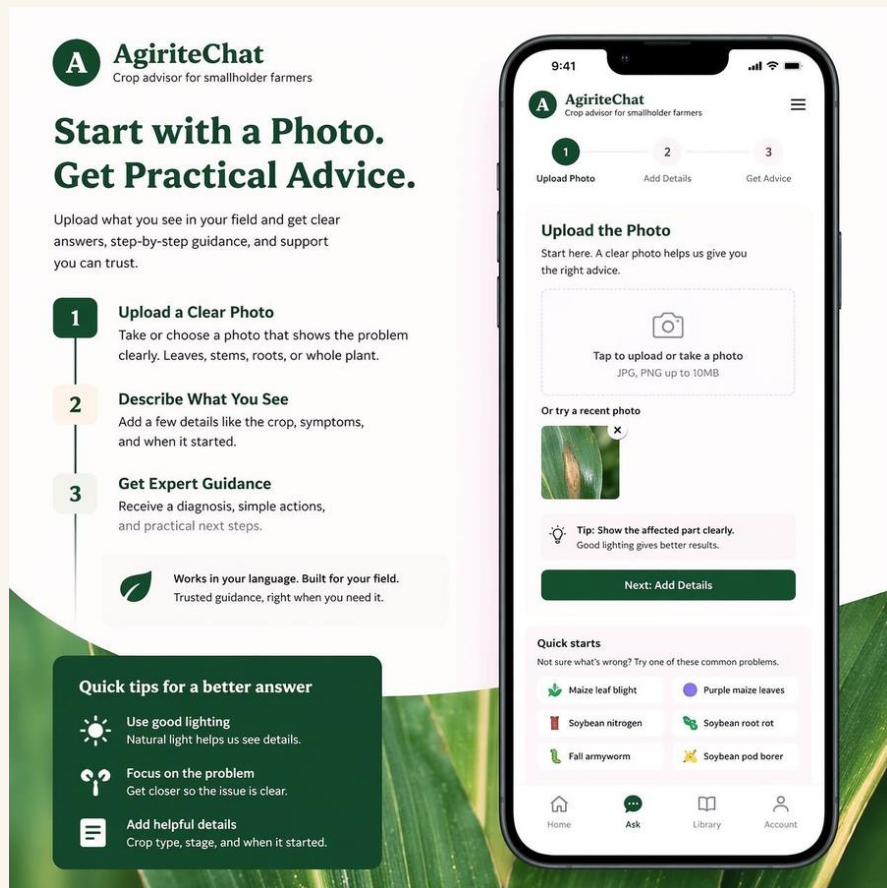
Green = KB. Amber = AI generated.



Architecture & tools

Text LLM	Llama 3.3 70B via Groq
Vision	Llama 4 Scout 17B via Groq
Voice	Whisper v3 Turbo via Groq
Embeddings	all-MiniLM-L6-v2 (local)
Agent	LangGraph · 7-node state machine
Retrieval	NumPy cosine similarity
Weather	Open-Meteo (free, no key)
KB	101 entries · 98 sources
Hosting	Streamlit Cloud (free)

Total: ~3,400 lines · 101 KB entries · 98 sources · 4 languages · \$0 cost



Safety by design, not afterthought



Never invents information

Constrained to KB entries only



Never provides doses

Refers to extension for quantities



Refuses when uncertain

Honest "I don't know" over guessing



Source separation

Green (KB) vs amber (AI) badges



Vision describes, doesn't diagnose

Prevents hallucination



Extension referral always present

"When to seek support" in every answer



Transparent confidence scores

Farmers see how sure the system is

Thank you

agiritechat.streamlit.app

github.com/Jean-ClaudeN/AgiriteChat

*An agent is defined
by knowing when not to act.*

— Design philosophy

